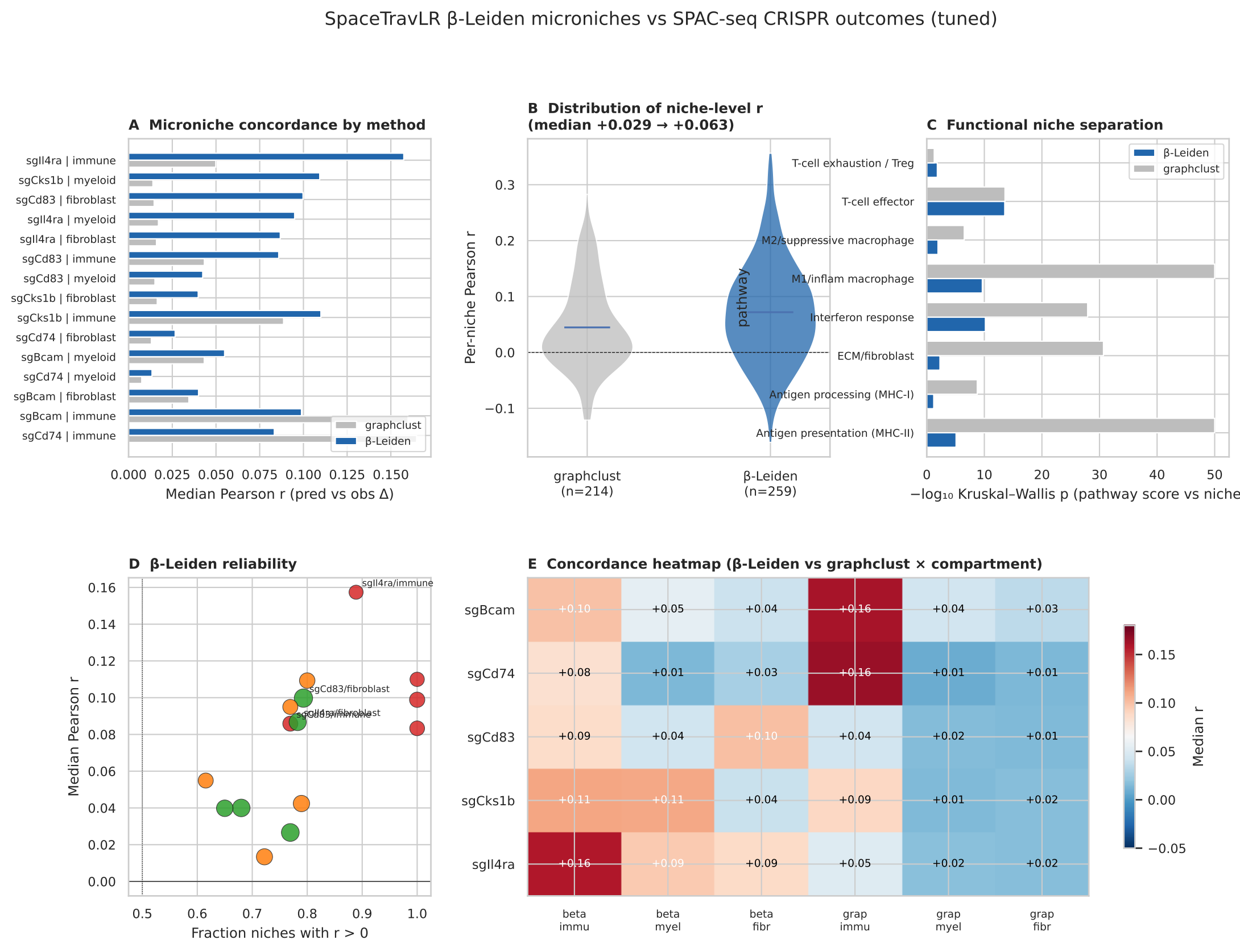
SpaceTravLR β-Leiden microniches vs SPAC-seq CRISPR outcomes (tuned)
A Microniche concordance by method
sgIl4ra | immune
sgCks1b | myeloid
sgCd83 | fibroblast
sgIl4ra | myeloid
sgIl4ra | fibroblast
sgCd83 | immune
sgCd83 | myeloid
sgCks1b | fibroblast
sgCks1b | immune
sgCd74 | fibroblast
sgBcam | myeloid
sgCd74 | myeloid
sgBcam | fibroblast
sgBcam | immune
sgCd74 | immune
graphclust
β-Leiden
0.000 0.025 0.050 0.075 0.100 0.125 0.150
Median Pearson r (pred vs obs Δ)
B Distribution of niche-level r
(median +0.029 → +0.063)
Per-niche Pearson r
0.3
0.2
0.1
0.0
−0.1
pathway
T-cell exhaustion / Treg
T-cell effector
M2/suppressive macrophage
M1/inflam macrophage
Interferon response
ECM/fibroblast
Antigen processing (MHC-I)
Antigen presentation (MHC-II)
graphclust
(n=214)
β-Leiden
(n=259)
C Functional niche separation
β-Leiden
graphclust
0 10 20 30 40 50
−log₁₀ Kruskal–Wallis p (pathway score vs niche
D β-Leiden reliability
Median Pearson r
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
sgIl4ra/immune
sgCd83/fibroblast
sgIl4ra/fibroblast
sgCd83/immune
0.5 0.6 0.7 0.8 0.9 1.0
Fraction niches with r > 0
E Concordance heatmap (β-Leiden vs graphclust × compartment)
sgBcam +0.10 +0.05 +0.04 +0.16 +0.04 +0.03
sgCd74 +0.08 +0.01 +0.03 +0.16 +0.01 +0.01
sgCd83 +0.09 +0.04 +0.10 +0.04 +0.02 +0.01
sgCks1b +0.11 +0.11 +0.04 +0.09 +0.01 +0.02
sgIl4ra +0.16 +0.09 +0.09 +0.05 +0.02 +0.02
beta immu
beta myel
beta fibr
grap immu
grap myel
grap fibr
0.15
0.10
0.05
0.00
−0.05
Median r